

INARI KIRAKKAJARVI, Finland

WMO: 028580

Lat: 69.582N

Lon: 28.894E

Elev: 106

StdP: 100.06

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.1	(c) -28.8	(d) -35.4	(e) 0.1	(f) -32.1	(g) -31.7	(h) 0.2	(i) -28.7	(j) 8.9	(k) -5.3	(l) 7.8	(m) -5.7	(n) 1.0	(o) 250	(p) 0.757

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	23.9	16.0	21.2	15.2	19.0	14.1	17.2	22.4	15.8	20.2	14.6	18.3	3.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	10.8	18.7	13.7	9.9	17.2	12.7	9.3	16.1	48.7	22.5	44.5	20.5	41.2	18.5	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.6	6.6	5.9	-35.6	27.2	2.6	2.5	-37.4	29.0	-38.9	30.5	-40.3	31.9	-42.2	33.8
			-34.5	18.5	4.3	1.9	-37.6	19.9	-40.1	20.9	-42.5	22.0	-45.7	23.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.4	-13.6	-11.7	-7.3	-1.6	4.3	8.6	13.0	11.2	7.6	0.1	-6.4	-9.5		
	DBStd	10.31	7.78	7.66	5.63	4.32	3.73	3.74	4.11	3.37	3.27	3.99	5.50	6.71		
	HDD10.0	4000	731	608	536	347	181	71	16	24	87	306	491	604		
	HDD18.3	6833	989	842	794	597	434	293	172	222	323	564	741	862		
	CDD10.0	218	0	0	0	0	5	28	109	62	13	0	0	0		
	CDD18.3	8	0	0	0	0	0	0	7	1	0	0	0	0		
	CDH23.3	107	0	0	0	0	1	3	91	13	0	0	0	0		
	CDH26.7	24	0	0	0	0	0	0	20	4	0	0	0	0		
(14) Wind	WSAvg	2.9	2.5	2.6	3.1	3.0	3.0	3.5	3.1	2.8	2.9	2.8	2.5	2.6		
(15) Precipitation	PrecAvg	453	31	27	24	23	28	50	59	60	38	45	33	32		
	PrecMax	613	54	53	50	67	54	120	125	175	66	99	58	62		
	PrecMin	347	13	9	10	3	10	9	16	20	10	18	9	17		
	PrecStd	65	11	12	11	14	11	30	30	35	15	22	14	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.1	2.7	5.1	10.2	20.0	22.5	29.5	24.6	18.2	10.1	3.9	3.7		
		MCWB	1.2	1.5	2.5	5.8	11.9	13.2	18.1	17.3	14.3	8.0	2.8	2.1		
	2%	DB	-0.4	1.3	2.9	7.2	16.3	19.9	26.3	21.8	16.1	8.7	1.9	1.4		
		MCWB	-1.1	0.3	1.0	3.7	9.9	13.0	16.8	16.0	12.2	7.2	1.1	0.7		
	5%	DB	-2.1	0.0	1.6	5.5	13.2	17.5	23.0	19.3	14.6	6.9	1.0	0.2		
		MCWB	-2.5	-0.8	0.1	2.7	8.7	12.1	15.5	14.8	11.6	5.8	0.5	-0.3		
	10%	DB	-4.0	-1.5	0.4	4.1	10.7	15.0	20.5	17.2	12.8	5.3	0.2	-0.8		
		MCWB	-4.5	-2.3	-0.8	1.8	6.8	10.6	15.1	13.8	10.6	4.5	-0.3	-1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.3	2.0	2.5	6.0	13.2	15.6	19.5	19.0	14.6	8.3	2.8	2.1		
		MCDB	2.1	2.5	4.5	9.8	19.2	19.7	23.7	22.7	17.9	9.5	3.9	3.2		
	2%	WB	-1.0	0.3	1.2	4.1	10.4	13.7	17.9	16.7	13.1	7.3	1.4	0.8		
		MCDB	-0.3	1.1	2.4	6.5	15.4	17.5	24.9	20.7	15.4	8.7	1.9	1.2		
	5%	WB	-2.6	-0.9	0.3	2.9	8.7	12.6	16.5	15.2	11.7	6.0	0.6	-0.2		
		MCDB	-2.1	-0.1	1.4	5.0	12.6	16.8	21.9	18.6	13.8	6.9	0.9	0.2		
	10%	WB	-4.3	-2.3	-0.7	2.1	7.2	11.2	15.3	14.1	10.7	4.6	-0.2	-1.3		
		MCDB	-4.0	-1.6	0.4	3.8	10.2	15.0	19.9	16.8	12.5	5.2	0.1	-0.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	7.7	9.1	7.9	7.5	6.9	7.9	7.2	6.2	4.5	6.1	7.9		
		MCDBR	8.6	5.4	6.1	8.3	13.3	11.1	13.2	10.9	9.3	4.6	4.2	5.9		
	5% WB	MCWBR	8.4	4.9	4.9	5.5	7.9	5.6	5.7	5.9	5.8	3.8	4.0	5.7		
		MCWBR	8.8	5.4	5.7	7.0	11.7	10.3	11.5	9.9	8.1	4.5	4.4	5.9		
(40) Clear-Sky Solar Irradiance	taub		0.202	0.249	0.295	0.325	0.349	0.346	0.360	0.342	0.308	0.264	N/A	N/A		
	taud		1.906	2.038	2.118	2.182	2.346	2.459	2.446	2.475	2.529	2.221	N/A	N/A		
	Ebn at Noon		19	474	700	803	832	850	821	794	728	465	0	0		
	Edh at Noon		4	43	80	103	100	92	91	78	56	35	0	0		
(44) All-Sky Solar Radiation	RadAvg		0.01	0.35	1.70	3.47	4.51	4.80	4.38	3.02	1.61	0.48	0.00	0.00		
	RadStd		0.00	0.04	0.10	0.25	0.45	0.42	0.47	0.30	0.12	0.06	0.00	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (19 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page